Revision of MSK Content PT4000 and PT7003 Semester 1: Autumn

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Module's Learning outcomes (Passing criteria)

- Planning and performing an effective and full physiotherapy Assessment of various MSK, neuro, and CRP conditions. Further, understanding the underlying reason of the need to collect information.
- Prioritisation of the information that are needed to collect (what must be performed first, what should be performed then and what could be performed).
- Analyse information and clinically reason information collected to reach a diagnosis of the condition of the patient while appreciating the biopsychosocial model approach.
- Creating a person-centred clear problem list and inferring and planning treatments to solve these problems.
- Planning and applying patient-centred treatment plan including therapeutic exercises, manual therapy and health promotion.
- Effectively coach the patient, take charge of the session and guide their recovery journey.
- Safely clearing red flags and providing safety netting information to the patients

Taught Curriculum MSK (in-order of teaching)

- Introduction to PT4000 module & concept of long-term care
- MSK: Introduction to musculoskeletal conditions (and OA)
- MSK: Touch in clinical setting (practical)
- Introduction to PT7004 module & acute and emergency physiotherapy
- MSK: Therapeutic handling skills (practical)
- MSK: Clinical muscle testing (practical)
- SDL 1: acute and emergency care pathways
- SDL 1: Concepts of long-term care - an introduction (the long-term plan, commissioning/provider landscape)
- MSK: Introduction to MSK assessment
- MSK: Clinical Reasoning and diagnosis in MSK
- MSK: Discussion of muscle injuries epidemiology and pre-disposing factors
- MSK: Principle of assessments in acute MSK injuries
- MSK: Muscle injury (case study)
- SDL2: Knee OA (case study)
- MSK: Musculoskeletal Assessment (Subjective assessment)
- MSK: Discussion of Tendon Injuries
- MSK: Introduction to MSK Treatment
- MSK: Tendon (case study)
- MSK: Principles of movement analysis and Range of Motion Assessment (practical)
- MSK: Discussion of ligament injuries
- MSK: Ligament (case study)
- Knee OA Assessment (Case Study)
- MSK: Hip Assessment (practical)
- MSK: Knee assessment (Practical)
- MSK: Bone injuries and management (fracture management and basic orthopaedics)
- MSK: Mobility Aids (practical)
- SDL 2: Fracture management (case study)
- MSK: Principles of exercise 1: Types of exercises, Specificity and training requirements - strength training
- MSK: Exercise prescription - post fracture orthopaedics procedures
- SDL 3: Post fracture and orthopaedics case study
- MSK: Principles of exercise 2: Flexibility and Stretching
- MSK: Manual Therapy - Lower Limb (practical)
- MSK: Ankle Assessment (practical)
- MSK: Gait Analysis (Practical)
- MSK: Foot assessment (practical)
- MSK: Knee TTT (Case Study)
- MSK: Flexibility and Stretching & Treatment
- MSK: Neuro integrity Assessment (practical)
- MSK: Principles of Exercise 4: Physical activity and health promotions
- MSK: Electrophysical agents for acute conditions - TENS, CRYOTHERAPY & ULTRASOUND
- MSK: Neurodynamic Assessment (practical)
- Christmas Holiday
- Examination
- Revision

Taught curriculum (types of sessions)

Lectures

- Introduction to PT4000 module & concept of long-term care
- MSK: Introduction to musculoskeletal conditions (and OA)
- Introduction to PT7004 module & acute and emergency physiotherapy
- MSK: Introduction to MSK assessment
- MSK: Clinical Reasoning and diagnosis in MSK
- MSK: Discussion of muscle injuries epidemiology and pre-disposing factors
- MSK: Principle of assessments in acute MSK injuries
- MSK: Musculoskeletal Assessment (Subjective assessment)
- MSK: Discussion of Tendon Injuries
- MSK: Discussion of ligament injuries
- MSK: Introduction to MSK Treatment
- MSK: Bone injuries and management (fracture management and basic orthopaedics)
- MSK: Principles of exercise 1: Types of exercises, Specificity and training requirements - strength training
- MSK: Exercise prescription - post fracture orthopaedics procedures
- MSK: Principles of exercise 2: Flexibility and Stretching
- MSK: Flexibility and Stretching & Treatment
- MSK: Principles of Exercise 4: Physical activity and health promotions
- MSK: Electrophysical agents for acute conditions - TENS, Cryotherapy & Ultrasound.

Practical

- MSK: Touch in clinical setting (practical)
- MSK: Therapeutic handling skills (practical)
- MSK: Clinical muscle testing (practical)
- MSK: Principles of movement analysis and Range of Motion Assessment (practical)
- MSK: Hip Assessment (practical)
- MSK: Knee assessment (Practical)
- MSK: Mobility Aids (practical)
- MSK: Manual Therapy - Lower Limb (practical)
- MSK: Ankle Assessment (practical)
- MSK: Gait Analysis (Practical)
- MSK: Foot assessment (practical)
- MSK: Neuro integrity Assessment (practical)
- MSK: Neurodynamic Assessment (practical)

Seminars

- MSK: Muscle injury (case study)
- MSK: Tendon (case study)
- MSK: Ligament (case study)
- Knee OA Assessment (Case Study)
- MSK: Knee TTT (Case Study)

SDL

- SDL 1: acute and emergency care pathways
- SDL 1: Concepts of long-term care - an introduction (the long-term plan, commissioning/provider landscape)
- SDL2: Knee OA (case study)
- SDL 2: Fracture management (case study)
- SDL 3: Post fracture and orthopaedics case study

Expectations from Semester 1

1. You can effectively perform a full MSK assessment for **various conditions** of the lower limb including **chronic and acute.**
2. You can interpret and reason the findings of your assessment .
3. You can create the problems of the patient and infer the solutions that you could offer as a physiotherapy student.
4. You can clear red flags and work safely with the patient for the conditions of the lower Limb (fractures, dislocations, sprains, strains…etc).
5. You can apply treatments that target the patient issues and offer them relief.
6. You can follow up with the patient and progress or regress treatments.
7. You can effectively coach the patient, take charge of the session and guide their recovery journey.
8. You can effectively communicate, and safety net the patient to potential worsening of symptoms.

Examples of conditions of LL

1. Hip Joint

Acute Conditions

- Hip dislocation
- Labral tears
- Fractures (e.g. neck of femur, shaft of femur, tibial condyles…etc)
- Muscle strains (e.g. Hamstring, adductors…etc)
- Hamstring avulsion injuries
- Acute groin strain

Chronic Conditions

- Osteoarthritis
- Femoroacetabular impingement (FAI)
- Tendinopathies (e.g., gluteus medius or iliopsoas tendinopathy…etc)
- Snapping hip syndrome (internal or external causes)
- Bursitis (Trochanteric, Ischial, iliopsoas)
- Pubic symphysis dysfunction
- Chronic hamstring tendinopathy

2. Knee Joint

Acute Conditions

- Ligament injuries (ACL, PCL, MCL, or LCL sprains/ruptures)
- Meniscal injuries (e.g., bucket-handle tears)
- Patellar dislocation or subluxation
- Fractures (e.g., patellar fractures, tibial plateau fractures)
- Quadriceps or patellar tendon rupture
- Acute impingement of Hoffa's fat pad
- Bursitis (Prepatellar, suprapatellar, infrapatellar, pes anserine)

Chronic Conditions

- Osteoarthritis
- Patellofemoral pain syndrome (PFPS)
- Patellar maltracking
- Chondromalacia patellae
- Iliotibial band (ITB) syndrome
- Tendinopathies (Patellar, Quadriceps)
- Baker’s cyst
- Hoffa’s fat pad syndrome
- Gout or pseudogout

3- Ankle Joint and Foot

Acute Conditions

- Fractures (Lateral or medial malleolus fractures, Talus or calcaneus fractures)
- Metatarsal and foot fractures (e.g., stress fractures, Lisfranc injuries)
- Ligament sprains (ATFL injury, high ankle sprain)
- Achilles tendon rupture
- Ankle dislocation

Chronic Conditions

- Chronic ankle instability
- Tendinopathy (Achilles, Peroneal)
- Posterior tibial tendon dysfunction (PTTD)
- Osteoarthritis of the ankle
- Rheumatoid arthritis of the toes.
- Tarsal tunnel syndrome
- Anterior impingement syndrome
- Plantar fasciitis
- Morton’s neuroma and Metatarsalgia
- Bunions (hallux valgus)
- Stress fractures (e.g., navicular, metatarsal bones)
- Flatfoot (pes planus) or high arches (pes cavus) deformities

Assignment for the second semester

ASSIGNMENT PT4000, PT7003

Summary of Musculoskeletal (MSK) Conditions

In your groups, work together to prepare a detailed document summarising the evidence-based musculoskeletal conditions affecting the body regions assigned to your group, as outlined in the table below:

Groups	BSc	MSc
Group A	Shoulder joint	Ankle and foot joints
Group B	Elbow joint	Cervical spine
Group C	Wrist and hand joints	Thoracic spine
Group D	Hip joint	Lumbar spine
Group E	Knee joint	Sacroiliac joint (SIJ)

Document Requirements

- **Cover page** including
  - The body region
  - The name of the assignment
  - The names of the students who contributed to the work, with asterisk next to the groups leader (s)
- **Body of the document:**
  - For each assigned body region, include the following details:

Condition	Provide the name of the condition being discussed
Pathophysiology	Explain the underlying mechanisms and causes of the condition
Signs and symptoms	List the clinical features commonly observed
Clusters of tests for diagnose	Identify diagnostic tests or clusters of tests that help confirm the condition
Differential Diagnosis	List conditions that may resemble the case. Explain the tests used to differentiate it from other conditions.
Management	Guidelines advice of what would help this patient recover.
References	Including the references used to summarise your work

Students Contribution

- Each group will nominate a leader who will be responsible of the curation of the students' contributions, add them to one document.
- At the end of your document, include a table showing the contribution of each group member to the project.

Example:

Leader of the Group	Name of the student
Student 1 name	Conditions A,B,C…
Student 2 name	Conditions H,I,G…
Student 3 name	Conditions L, M,N…
Submission Ambassador	Name of the student who responsible of the submission on time.

Submission and Evaluation

Your document will be evaluated on



Content

Comprehensive coverage of conditions.



Accuracy

accurate evidence-based information with references.



Teamwork

Clear contribution from all group members.



Presentation

Clear, concise, and well-organized document.

- **Submission of a collated word document from each group will be submitted**
- BSc 30/03/2025
- MSc 15/04/2025
- **Presentations:** TBC

Example

Condition	Rotator Cuff Tear
Pathophysiology	A rotator cuff tear involves damage to one or more of the four rotator cuff tendons (supraspinatus, infraspinatus, teres minor, and subscapularis). Common causes include acute trauma, repetitive microtrauma, or age-related degenerative changes. Degenerative tears are most prevalent in individuals over 50 years old, with prevalence increasing with age (Yamamoto et al., 2010).
Signs and symptoms	Pain, especially during overhead activities or at night. Weakness in abduction and external rotation. Limited range of motion (ROM). Crepitus or clicking sounds during movement.
Clusters of tests for diagnose	Evidence suggests using a combination of tests for greater diagnostic accuracy (Hegedus et al., 2012). 1. Painful Arc Test (Sensitivity: 74%, Specificity: 81%). 2. Infraspinatus Strength Test (Sensitivity: 62%, Specificity: 73%). 3. Drop Arm Test (Sensitivity: 44%, Specificity: 92%). Note: A positive result on all three tests strongly indicates a rotator cuff tear (Park et al., 2005).
Differential Diagnosis	1. Adhesive Capsulitis: Marked restriction in both active and passive ROM, particularly external rotation. 2. Subacromial Impingement Syndrome (SIS): Pain during the painful arc test but without significant weakness. Positive Hawkins-Kennedy Test combined with a normal Drop Arm Test supports SIS over a tear. 3. Cervical Radiculopathy: Neck pain radiating to the shoulder, with associated paraesthesia or weakness in the arm. Spurling's Test and imaging (MRI/CT) for confirmation.
Management	Advise rest in the acute phase. Following this, activity should gradually be increased. Advise modification of activities that exacerbate symptoms (such as reaching overhead). Usual activities, within the limits of pain, should be restarted as soon as possible. Physiotherapy may include postural correction, motor control retraining, stretching and strengthening of the rotator cuff and scapular muscles, and manual therapy. Low intensity and high-frequency exercise, that is within the person's pain threshold, should be performed. Physiotherapy can be continued for a further 6 weeks if there is improvement in the person's symptoms or function (Nice Guidelines, 2022).
References	Hegedus, E. J., Cook, C., Wright, A., et al. (2012). Diagnostic accuracy of clinical tests of the shoulder: a systematic review and meta-analysis. BMJ Open. Yamamoto, A., Takagishi, K., Kobayashi, T., et al. (2010). Prevalence and risk factors of a rotator cuff tear in the general population. J Shoulder Elbow Surg. Kuhn, J. E., Dunn, W. R., Sanders, R., et al. (2013). Park, H. B., Yokota, A., Gill, H. S., & El Rassi, G. (2005). Diagnostic accuracy of clinical tests for the different degrees of subacromial impingement syndrome and rotator cuff tears. J Bone Joint Surg. Am.Effectiveness of physical therapy in treating atraumatic full-thickness rotator cuff tears: a multicenter prospective cohort study. J Shoulder Elbow Surg. Nice Guidelines, Rotator Cuff disorders, 2022.

Questions

